# THE RIGHT ADJOINT OF PUSHFORWARD FOR A PROPER FLAT FINITE PRESENTATION MORPHISM COMMUTES WITH DIRECT SUMS

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0E4I (sources/stacks-duality/pinned-original-source.tex line 2831).

*Allowed standard prerequisites (editorial).*

- The pushforward of a perfect object along a flat proper morphism is perfect. Derived Categories of Schemes, Lemma labelled perfect-lemma-flat-proper-perfect-direct-image-general, cited at proof lines 2865-2868. Makes the Hom computations finite.
- Perfect objects are compact. Derived Categories of Schemes, Proposition labelled perfect-proposition-compact-is-perfect, cited at proof lines 2876-2878. Turns Hom into a direct sum of Homs.
- Existence of a perfect generator of the quasi-coherent derived category. Derived Categories of Schemes, Theorem labelled perfect-theorem-bondal-van-den-Bergh, cited at proof lines 2880-2882. Concludes that the comparison map is an isomorphism.

*Source reference.* Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0B91: sources/stacks-chow/tags.txt, source lines 14159–14159. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2843 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original proposition 09M1: sources/stacks-chow/tags.txt, source lines 12164–12164. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2857 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original theorem 09IS: sources/stacks-chow/tags.txt, source lines 12050–12050. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2860 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *Let $Y$ be a quasi-compact and quasi-separated scheme. Let $f : X \to Y$ be a morphism of schemes which is proper, flat, and of finite presentation. Let $a$ be the right adjoint for $Rf_* : D_{QCoh}(\mathcal{O}_X) \to D_{QCoh}(\mathcal{O}_Y)$ of Lemma 0A9E. Then $a$ commutes with direct sums.*

*Proof.* Let $P$ be a perfect object of $D(\mathcal{O}_X)$. By Derived Categories of Schemes, Lemma 0B91 the complex $Rf_*P$ is perfect on $Y$. Let $K_i$ be a family of objects of $D_{QCoh}(\mathcal{O}_Y)$. Then

$$\begin{aligned} \mathrm{Hom}_{D(\mathcal{O}_X)}(P, a(\bigoplus K_i)) &= \mathrm{Hom}_{D(\mathcal{O}_Y)}(Rf_*P, \bigoplus K_i) \\ &= \bigoplus \mathrm{Hom}_{D(\mathcal{O}_Y)}(Rf_*P, K_i) \\ &= \bigoplus \mathrm{Hom}_{D(\mathcal{O}_X)}(P, a(K_i)) \end{aligned}$$

because a perfect object is compact (Derived Categories of Schemes, Proposition 09M1). Since $D_{QCoh}(\mathcal{O}_X)$ has a perfect generator (Derived Categories of Schemes, Theorem 09IS) we conclude that the map $\bigoplus a(K_i) \to a(\bigoplus K_i)$ is an isomorphism, i.e., $a$ commutes with direct sums. □